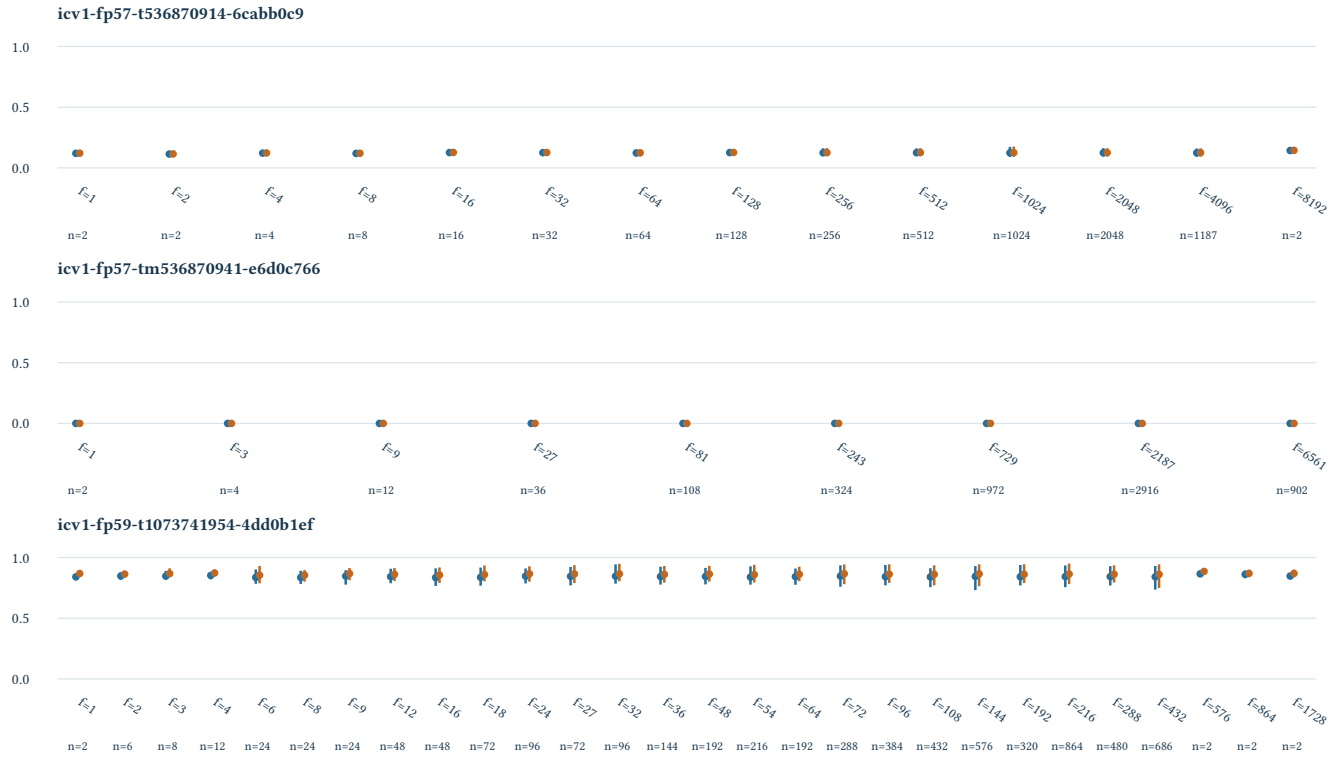
# Repaired volcano survey checkpoint

Run 20261018T2313Z-repaired-7937. 7,937 audited vertices, 126,992 accepted attempts.

## Repaired survey: all 51 stratum samples, projected size-64 MITM

Blue: equations/source trial. Orange: equations/actual oracle call. Native means and observed min–max; not confidence intervals.

Ordinal conductor axis; each point is a stratum/outcome group. n counts target attempts, not independent vertices.




Checkpoint 20261018T2313Z-repaired-7937: 7937 / 24823 vertices. All strata sampled; full-vertex distributions incomplete.

| Graph | Audited vertices | Planned vertices | Audited strata | Planned strata |
|---|---|---|---|---|
| mixed | 2656 | 10070 | 28 | 28 |
| three | 2638 | 6561 | 9 | 9 |
| two | 2643 | 8192 | 14 | 14 |

Observed outcomes: 81,666 verified recoveries, 39,550 exhausted attempts, 5,776 timeouts. Native source trials 202,646,648; actual decomposition calls 193,153,057; equations 4,117,173.

These pooled counts span different subgroups/bases and are accounting totals, not a common success probability. Multiple outcomes and all missing/censored observations remain distinct. Stratum samples are not complete populations.

Earlier interrupted attempts remain preserved and excluded from accepted means; their termination causes were not observed. The current checkpoint's context note identifies its stop or recovery. No sealed session, binary, queue or mathematical budget was changed.

The native inventory checks record seals, receipt/file hashes, ledger/queue identities and exact existing successful/exhausted replay coverage. This is receipt binding, not fresh or independent solver replay. The entire stopped campaign, including the unaccepted interruption, accompanies the inventory.

## Stratum distributions: recorded size-64 MITM

Trial yield includes repeated queries skipped before the oracle; actual-call yield excludes them. Identity skips occur before trial increment. Zero or unknown denominators have no yield. Timeout final counters are unknown, not zero. GAE with unpriced work is a lower bound; exhaustion costs an attempt, not a successful DLP. Native summary/CSV retain all other arms, phases and present-value n. The tables below use only recorded size-64 MITM reports.

### icv1-fp57-t536870914-6cabb0c9

| Conductor | Outcome | n | GAE min | GAE mean | GAE max | Mean/trial | Mean/call |
|---|---|---|---|---|---|---|---|
| 1 | verified | 2 | 19,530 | 19,970.0 | 20,410 | 0.119769 | 0.121079 |
| 2 | verified | 2 | 19,505 | 19,546.5 | 19,588 | 0.114059 | 0.114919 |
| 4 | verified | 4 | 19,466 | 20,048.5 | 20,714 | 0.121453 | 0.122819 |
| 8 | verified | 8 | 19,467 | 19,888.2 | 20,857 | 0.119064 | 0.120233 |
| 16 | verified | 16 | 19,210 | 19,796.4 | 20,524 | 0.126336 | 0.127545 |
| 32 | verified | 32 | 19,063 | 19,777.5 | 20,954 | 0.125379 | 0.126577 |
| 64 | verified | 64 | 18,937 | 19,903.8 | 21,779 | 0.122785 | 0.123939 |
| 128 | verified | 128 | 18,907 | 19,764.5 | 21,045 | 0.126269 | 0.127420 |
| 256 | verified | 256 | 18,833 | 19,875.1 | 21,476 | 0.124160 | 0.125334 |
| 512 | verified | 512 | 18,763 | 19,854.7 | 21,859 | 0.124704 | 0.125875 |
| 1024 | verified | 1024 | 18,691 | 19,864.5 | 22,854 | 0.124112 | 0.125262 |
| 2048 | verified | 2048 | 18,709 | 19,843.0 | 22,184 | 0.124210 | 0.125354 |
| 4096 | verified | 1187 | 18,733 | 19,836.8 | 22,060 | 0.124003 | 0.125155 |
| 8192 | verified | 2 | 19,369 | 19,846.5 | 20,324 | 0.143079 | 0.144412 |

### icv1-fp57-tm536870941-e6d0c766

| Conductor | Outcome | n | GAE min | GAE mean | GAE max | Mean/trial | Mean/call |
|---|---|---|---|---|---|---|---|
| 1 | exhausted | 2 | 20,239 | 20,252.0 | 20,265 | 0.000000 | 0.000000 |
| 3 | exhausted | 4 | 20,245 | 20,270.0 | 20,298 | 0.000000 | 0.000000 |
| 9 | exhausted | 12 | 20,240 | 20,273.7 | 20,327 | 0.000000 | 0.000000 |
| 27 | exhausted | 36 | 20,226 | 20,271.4 | 20,316 | 0.000000 | 0.000000 |
| 81 | exhausted | 108 | 20,232 | 20,278.6 | 20,340 | 0.000000 | 0.000000 |
| 243 | exhausted | 324 | 20,198 | 20,274.7 | 20,342 | 0.000000 | 0.000000 |
| 729 | exhausted | 972 | 20,205 | 20,275.2 | 20,337 | 0.000000 | 0.000000 |
| 2187 | exhausted | 2916 | 20,195 | 20,274.7 | 20,345 | 0.000000 | 0.000000 |
| 6561 | exhausted | 902 | 20,203 | 20,274.8 | 20,337 | 0.000000 | 0.000000 |

### icv1-fp59-t1073741954-4dd0b1ef

| Conductor | Outcome | n | GAE min | GAE mean | GAE max | Mean/trial | Mean/call |
|---|---|---|---|---|---|---|---|
| 1 | verified | 2 | 20,463 | 20,467.5 | 20,472 | 0.840387 | 0.868095 |
| 2 | verified | 6 | 20,012 | 20,126.2 | 20,241 | 0.847810 | 0.863518 |
| 3 | verified | 8 | 20,024 | 20,270.8 | 20,595 | 0.846633 | 0.867332 |
| 4 | verified | 12 | 19,899 | 20,210.8 | 20,534 | 0.852565 | 0.872370 |
| 6 | verified | 24 | 19,899 | 20,142.7 | 20,550 | 0.835873 | 0.854372 |
| 8 | verified | 24 | 19,807 | 20,142.4 | 20,648 | 0.835348 | 0.854639 |
| 9 | verified | 24 | 19,842 | 20,207.2 | 20,878 | 0.847141 | 0.867336 |
| 12 | verified | 48 | 19,783 | 20,183.1 | 20,749 | 0.842426 | 0.861242 |
| 16 | verified | 48 | 19,796 | 20,210.0 | 20,727 | 0.834727 | 0.855684 |
| 18 | verified | 72 | 19,863 | 20,202.9 | 20,836 | 0.837101 | 0.859458 |
| 24 | verified | 96 | 19,835 | 20,193.0 | 21,445 | 0.846245 | 0.865569 |
| 27 | verified | 72 | 19,786 | 20,146.4 | 20,934 | 0.844967 | 0.864468 |
| 32 | verified | 96 | 19,832 | 20,191.3 | 20,771 | 0.846372 | 0.865296 |
| 36 | verified | 144 | 19,723 | 20,156.5 | 20,928 | 0.842766 | 0.860938 |
| 48 | verified | 192 | 19,734 | 20,208.9 | 21,208 | 0.843601 | 0.863686 |
| 54 | verified | 216 | 19,757 | 20,230.8 | 21,129 | 0.838622 | 0.858583 |
| 64 | verified | 192 | 19,803 | 20,185.6 | 21,106 | 0.842563 | 0.862218 |
| 72 | verified | 288 | 19,741 | 20,178.8 | 21,201 | 0.847042 | 0.866266 |
| 96 | verified | 384 | 19,776 | 20,211.2 | 21,028 | 0.842064 | 0.862771 |
| 108 | verified | 432 | 19,785 | 20,225.3 | 21,246 | 0.840016 | 0.861047 |
| 144 | verified | 576 | 19,782 | 20,206.2 | 21,130 | 0.844469 | 0.864875 |
| 192 | verified | 320 | 19,781 | 20,210.3 | 21,284 | 0.841434 | 0.861458 |
| 216 | verified | 864 | 19,774 | 20,207.8 | 21,529 | 0.842896 | 0.863666 |
| 288 | verified | 480 | 19,800 | 20,211.8 | 21,114 | 0.842327 | 0.862172 |
| 432 | verified | 686 | 19,773 | 20,214.2 | 21,303 | 0.840860 | 0.861098 |
| 576 | verified | 2 | 20,153 | 20,259.0 | 20,365 | 0.865327 | 0.885298 |
| 864 | verified | 2 | 19,900 | 19,942.0 | 19,984 | 0.861202 | 0.868679 |
| 1728 | verified | 2 | 20,215 | 20,372.5 | 20,530 | 0.847351 | 0.868859 |

Protocol: actual largest prime subgroups; two public targets/model, one alternating round, zero warmups, bases 16/64, raw-base control, MITM/subtraction, rho.negation, BSGS and rho A/A. 4,096 source trials, 3-second child watchdog. Cold setup/projection/verification charged; unpriced work remains explicit. The full eight-target/five-round/million-trial matrix remains unexecuted.

Frozen release runner 35d89073f4d666b094c213a1d31b0c819b63600123e067c111af3c947d7609a9. Source 2bf7700b7, unchanged queue and driver. Earlier dev99, stopped release70, control/pilot and original C finite-field diagnostic are separate datasets; repeated workloads and different binaries are never pooled here. No performance improvement follows from this checkpoint's larger sample count.

Wall observations are unqualified and descriptive. Every capsule retains its host/environment observation; any resumed runtime epoch must remain identifiable. No admitted runtime, external fastest-method, scaling or sub-rho claim is made. Exact representation registration, transported targets, composite PH, independent arithmetic replay, original A–E full graphs and large-extension finite-field backends remain open. Original C's r=7 k=1 diagnostic is separate. Complete graph topology is for the declared fields and finite prime degrees, not every other prime or infinitely many compositions.

## Current checkpoint and source handoff

PID 50943 stopped cleanly on the owned STOP marker at 2026-10-18T23:13:54.932000+00:00. 1,800 additional vertices were audited after the 6,137-vertex delivery. No new attempt was interrupted by this pause. Earlier interruptions remain. An upstream merge changed the repository driver copy by removing drop(emit). The actual frozen ROOT driver source, binaries, queue and mathematical budgets are unchanged and remain the campaign's only execution inputs. No new root library validation or source/binary equivalence for the upstream edit is claimed. Original deadline: 20 October 20:13:42 UTC; total campaign cap: 6 GiB.